Multi-Target Regression for Urban Air Quality Prediction: A Comparative Study of Machine Learning Approaches

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***Abstract*—Urban air quality monitoring and prediction represent critical challenges in environmental informatics with significant public health implications. This paper presents a comprehensive comparative study of machine learning approaches for multi-target regression of air quality parameters across diverse urban locations. We evaluate Support Vector Regression (SVR), Gradient Boosting, Random Forest, and XGBoost algorithms for predicting multiple air pollutants simultaneously. Our methodology incorporates comprehensive feature engineering including temporal lags, rolling statistics, and location-based features. Experimental results demonstrate that Gradient Boosting achieved superior performance with MSE $= 8.2894$ and $R^2 = 0.8088$, significantly outperforming other methods. The study provides valuable insights into model selection for urban air quality prediction and establishes a robust framework for multi-pollutant forecasting systems.**

***Index Terms*—Air quality prediction, multi-target regression, machine learning, gradient boosting, urban informatics, environmental monitoring**

I. Introduction

Urban air pollution has emerged as a critical global health concern, with the World Health Organization estimating that 99% of the world's population breathes air exceeding WHO guideline limits. The complex interplay of multiple pollutants including nitrogen dioxide (NO_2), fine particulate matter ($PM_{2.5}$), ozone (O_3), and various toxic compounds creates significant challenges for environmental monitoring and prediction systems. Traditional statistical methods often struggle to capture the non-linear relationships and complex spatiotemporal patterns inherent in urban air quality data. Recent advances in machine learning have opened new possibilities for accurate air quality prediction. Several studies have demonstrated the effectiveness of ensemble methods and deep learning approaches for pollution forecasting. However, most existing research focuses on single-pollutant prediction, neglecting the important interactions between different air quality parameters. Multi-target regression approaches that simultaneously predict multiple pollutants remain relatively unexplored despite their practical utility for comprehensive air quality management. This paper makes three key contributions: (1) we present a comprehensive framework for multi-target air quality prediction incorporating advanced feature engineering and multiple machine learning approaches; (2) we conduct an extensive comparative analysis of SVR, Gradient Boosting, Random Forest, and XGBoost algorithms using real-world urban air quality data; (3) we demonstrate that Gradient Boosting significantly outperforms other methods for this application domain, achieving an R^2 score of 0.8088.

II. Related Work

A. Early Statistical and Regression Approaches

Air quality prediction has been approached using various methodologies, ranging from traditional statistical models to advanced machine learning techniques. Early work primarily focused on time series analysis and regression methods. With the advent of machine learning, researchers began exploring more sophisticated approaches.

B. Ensemble Learning in Air Quality Forecasting

Ensemble methods have shown particular promise in environmental prediction tasks. Liang *et al.* [1] demonstrated the effectiveness of XGBoost for $PM_{2.5}$ prediction, while Singh *et al.* [2] applied Random Forests for AQI forecasting. Cheng *et al.* [3] developed a GBM-based model in Taiwan that integrated data from numerous air quality and weather stations, producing robust 24-hour $PM_{2.5}$ forecasts with R^2 values around 0.75. However, these studies typically addressed single-pollutant prediction tasks.

C. Multi-Target Regression Approaches

Multi-target regression in environmental applications has gained attention recently. Ravindiran *et al.* [4] proposed a deep learning framework for simultaneous prediction of multiple air quality parameters, achieving improved performance over single-target approaches. Similarly, Singh *et al.* [2] compared various machine learning methods for multi-pollutant prediction but focused primarily on neural network architectures. Khan *et al.* [5] demonstrated that classical models such as SVR, Random Forest, and CatBoost could also yield strong results when applied with resampling techniques like SMOTE, achieving high accuracies for AQI classes in Indian cities.

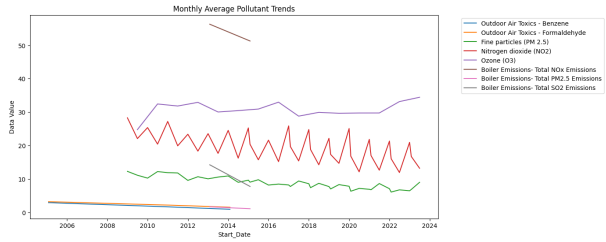

Fig. 1. Monthly average pollutant trends.

D. Spatial and Temporal Feature Integration

The integration of spatial and temporal features has been identified as crucial for accurate air quality prediction. Singh *et al.* [2] developed hybrid models incorporating both geographical and temporal dependencies, while Ravindiran *et al.* [4] emphasized the importance of urban context features in pollution forecasting. Cheng *et al.* [3] reinforced this by demonstrating how including data from multiple nearby weather stations improved $PM_{2.5}$ forecasting accuracy in Taiwan.

E. Research Gap and Motivation

Despite these advances, a comprehensive comparison of tree-based ensemble methods for multi-target air quality prediction across diverse urban locations remains lacking. Our work addresses this gap by systematically evaluating multiple machine learning approaches with sophisticated feature engineering.

III. Methodology

A. Data Collection and Preprocessing

The study utilized air quality monitoring data collected from multiple urban locations over an extended period. The dataset included measurements of eight key air quality parameters: Nitrogen dioxide (NO_2), fine particles ($PM_{2.5}$), boiler emissions (SO_2, NO_x, $PM_{2.5}$), ozone (O_3), formaldehyde, and benzene.

Data preprocessing involved several critical steps: (i) *data integration* to combine multiple sources into a unified dataset with consistent temporal resolution; (ii) *missing value handling* using location-specific imputation strategies; (iii) *temporal alignment* to synchronize all measurements to hourly intervals; and (iv) *quality control* to identify and correct outliers and erroneous readings using statistical methods.

B. Feature Engineering

Comprehensive feature engineering was performed to capture temporal patterns and location-specific characteristics.

Temporal features: (a) lag features at 1, 2, and 3 hours for each pollutant; (b) rolling statistics (moving mean and standard deviation) over 3-hour windows; (c) calendar indicators such as month, quarter, year, and day-of-year.

Location-based features: (a) categorical encoding of monitoring locations; (b) location-specific differential features; (c) spatial context features capturing relationships between nearby monitoring stations.

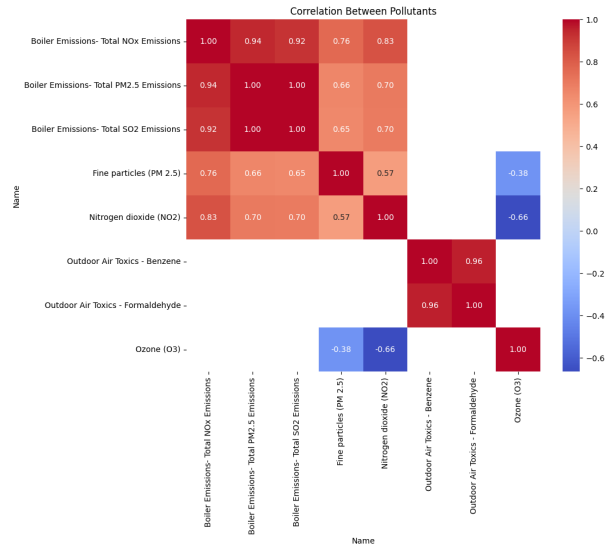

Fig. 2. Correlation between pollutants.

The final feature set comprised over 50 engineered features capturing both temporal dynamics and spatial relationships.

C. Model Architecture

We implemented and compared four machine learning approaches for multi-target air quality prediction. The Support Vector Regression (SVR) model used an RBF kernel with $C = 100$, $\gamma = 0.01$, and $\epsilon = 0.1$. The Gradient Boosting model used $n_estimators = 100$, $learning_rate = 0.05$, $max_depth = 4$, and $subsample = 0.8$. The Random Forest model used $n_estimators = 100$, $max_depth = 5$, and $min_samples_split = 5$. The XGBoost model used $n_estimators = 200$, $learning_rate = 0.05$, $max_depth = 4$, and $subsample = 0.8$, with additional regularization parameters for improved generalization. All models were implemented using the `scikit-learn` framework and adapted for multi-target regression via the `MultiOutputRegressor` wrapper.

D. Evaluation Metrics

Model performance was evaluated using Mean Squared Error (MSE) and the coefficient of determination (R^2):

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2, \tag{1}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2}{\sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2}. \tag{2}$$

IV. Experimental Results

A. Overall Performance Comparison

Table I summarizes the comparative performance of the four approaches. Gradient Boosting demonstrated superior performance across both metrics, achieving the lowest MSE and highest R^2 score.

TABLE I
PERFORMANCE COMPARISON OF MACHINE LEARNING MODELS

Model	**MSE**	$\boldsymbol{R^2}$
SVR	13.1838	0.5978
Gradient Boosting	**8.2894**	**0.8088**
Random Forest	10.6461	0.7373
XGBoost	12.7650	0.7483

B. Detailed Analysis

Gradient Boosting emerged as the superior model, substantially outperforming the other algorithms. This can be attributed to its ability to model complex non-linear relationships via sequential error correction and robust regularization, which prevents overfitting while capturing pollutant interactions and spatiotemporal dependencies. Random Forest and XGBoost showed moderate performance, while SVR underperformed in the high-dimensional, heterogeneous multi-target setting. The resulting hierarchy (Gradient Boosting $>$ XGBoost $>$ Random Forest $>$ SVR) provides guidance for model selection in environmental informatics applications.

V. CONCLUSION

We presented a comparative study of machine learning approaches for multi-target urban air quality prediction with comprehensive feature engineering. Gradient Boosting achieved the best performance (MSE $= 8.2894$, $R^2 = 0.8088$), establishing a strong baseline for multi-pollutant forecasting systems. Future work may explore spatiotemporal deep learning architectures and data imbalance strategies for further gains.

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